

CHEMICAL EQUILIBRIA IN METHANOL SYNTHESIS

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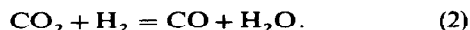
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Abstract—The chemical equilibria of the methanol reaction and the water–gas shift reaction, starting from carbon monoxide, carbon dioxide and hydrogen, were studied in a fixed-bed catalytic reactor at $P = 10$ – 80 bar and $T = 200$ – 270°C . It was found that the chemical equilibria could be described very well by thermochemical data based on ideal gas behaviour in combination with a correction for the non-ideality of the gas mixture as predicted by the Soave–Redlich–Kwong equation of state. This correction for non-ideality results in significantly better agreement with experimental data than a correction based on the original Redlich–Kwong equation of state, the Peng–Robinson equation of state, the virial equation truncated after the second virial coefficient, Lewis and Randall's rule, or not correcting at all for non-ideality, thus assuming ideal gas behaviour.

1. INTRODUCTION

Methanol, an important industrial chemical is produced on a large scale in the so-called “low-pressure” (50–100 bar) process. The formation of methanol is catalysed by Cu–Zn–Al or Cu–Zn–Cr mixed oxides. Important design factors in modelling a methanol reactor are the values of the equilibrium constants of the following reactions:



However, a thorough literature study indicates that no reliable choice between various chemical equilibrium literature data can be made. Literature data on the methanol equilibrium are numerous but conflicting. Literature data on the water–gas shift equilibrium are scarce in the temperature region of interest (i.e. 200–300°C). The necessary extrapolation makes these data unreliable too.

As part of the research programme in our laboratory to investigate possible types of methanol reactor, we studied the chemical equilibria of the reactions mentioned above. The question we ask ourselves is: Can the chemical equilibria be represented reliably by equilibrium constants which are calculated from thermochemical data assuming ideal gas behaviour and subsequently corrected for non-ideality using a suitable equation of state?

2. THEORY

The equilibrium constants of the methanol reaction (1) and the water–gas shift reaction (2) are defined as

follows:

$$K_{f_1} = \left(\frac{f_{\text{CH}_3\text{OH}}}{f_{\text{CO}} f_{\text{H}_2}^2} \right)_{\text{Eq}} = \left(\frac{\phi_{\text{CH}_3\text{OH}}}{\phi_{\text{CO}} \phi_{\text{H}_2}^2} \right)_{\text{Eq}} \left(\frac{P_{\text{CH}_3\text{OH}}}{P_{\text{CO}} P_{\text{H}_2}^2} \right)_{\text{Eq}} \quad (3)$$

$$K_{f_2} = \left(\frac{f_{\text{CO}} f_{\text{H}_2\text{O}}}{f_{\text{CO}_2} f_{\text{H}_2}} \right)_{\text{Eq}} = \left(\frac{\phi_{\text{CO}} \phi_{\text{H}_2\text{O}}}{\phi_{\text{CO}_2} \phi_{\text{H}_2}} \right)_{\text{Eq}} \left(\frac{P_{\text{CO}} P_{\text{H}_2\text{O}}}{P_{\text{CO}_2} P_{\text{H}_2}} \right)_{\text{Eq}} \quad (4)$$

Here f_i is the fugacity, ϕ_i is the fugacity coefficient and P_i is the partial pressure of component i . Since most gases show ideal gas behaviour at low pressures, we can make the following assumption at a pressure of 1 bar:

$$K_f = K_p^\circ. \quad (5)$$

Then K_p° is a function of temperature only. However, K_ϕ can be a function of temperature, pressure and composition.

The value of K_p° can be calculated from the following thermodynamic relationships:

$$-\Delta G^\circ(T) = RT \ln K_p^\circ(T) \quad (6)$$

$$\frac{-\Delta G^\circ(T)}{T} = \frac{-\Delta G^\circ(T_{\text{ref}})}{T_{\text{ref}}} - \int_{T_{\text{ref}}}^T \frac{\Delta H^\circ(T)}{T^2} dT \quad (7)$$

$$\Delta H^\circ(T) = \Delta H^\circ(T_{\text{ref}}) + \int_{T_{\text{ref}}}^T \Delta C_p^\circ(T) dT. \quad (8)$$

Since the specific heats of the components are known to be polynomial functions of T , the integrations in eqs (7) and (8) can easily be carried out. For example, when the polynomials are of third degree, the following function is obtained:

$$\ln K_p^\circ = \frac{1}{RT} (a_1 + a_2 T + a_3 T^2 + a_4 T^3 + a_5 T^4 + a_6 T \ln T). \quad (9)$$

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For relatively small temperature ranges this equation is unnecessarily complicated. A simplification which gives excellent results for the temperature range 200–300°C is:

$$\log_{10} K_p^\circ = \frac{b_1}{T} + b_2 \quad (10)$$

Using $\Delta H_f^\circ(25^\circ\text{C})$ and $\Delta G_f^\circ(25^\circ\text{C})$ values (Rossini *et al.*, 1961) in combination with specific heat data (Reid *et al.*, 1977), we obtain the following results:

$$\log_{10} K_{p_1}^\circ = \frac{5139}{T} - 12.621 \quad (\text{pressures in bars}) \quad (11)$$

$$\log_{10} K_{p_2}^\circ = \frac{-2073}{T} + 2.029. \quad (12)$$

It is possible to calculate K_ϕ values using an equation of state. For a pressure-explicit equation of state, the fugacity coefficient of a component i in a mixture is given by (Prausnitz, 1969):

$$\ln \phi_i = \frac{1}{RT} \int_V^\infty \left[\left(\frac{\partial P}{\partial n_i} \right)_{T,V,n_j} - \frac{RT}{V} \right] dV - \ln Z. \quad (13)$$

There are many equations of state available. The choice is not easily made as the system is a complicated one. It is a high-pressure system containing polar and non-polar molecules, some of which are above and some of which are below their critical temperatures. We have

decided to investigate which of the equations of state listed in Table 1 is the most suitable one for use under the conditions in a "low-pressure" methanol reactor.

The most well-known equation of state is of course the ideal gas law. This equation yields ϕ and K_ϕ values equal to 1.

Equations of state that are often used in practice are the so-called modifications of the original van der Waals equation (van der Waals, 1873), such as the Redlich–Kwong (Redlich and Kwong, 1949), the Soave–Redlich–Kwong (Soave, 1971) and the Peng–Robinson equation of state (Peng and Robinson, 1976). The parameters a and b in these equations can be calculated from the critical pressures and temperatures and the acentric factors of the components (in the case of the Redlich–Kwong equation of state, only critical pressures and temperatures are used). In methanol synthesis at least five components are present. Their critical properties and acentric factors are listed in Table 2.

The original mixing rules are used without interaction coefficients (Redlich and Kwong, 1949).

$$a = \sum_i \sum_j y_i y_j \sqrt{a_i a_j} \quad (14)$$

$$b = \sum_i y_i b_i. \quad (15)$$

An advantage of the virial equation in comparison with the van der Waals-like equations is its theoretical

Table 1. Equations of state evaluated in this study

Equation of state	Abbreviation	Formula $Z = Pv/RT =$
Ideal gas law	IGL	1
Redlich–Kwong equation of state	RK	$\frac{v}{v-b} - \frac{1}{RT^{3/2}(v+b)}$
Soave–Redlich–Kwong equation of state	RKS	$\frac{v}{v-b} - \frac{a(T)}{RT(v+b)}$
Peng–Robinson equation of state	PR	$\frac{v}{v-b} - \frac{a(T)v}{RT(v(v+b)+b(v-b))}$
Virial equation	Vir	$1 + \frac{B}{v} + \frac{C}{v^2} + \dots$

Table 2. Critical properties and acentric factors

Component	$P_{c,i}$ (bar)	$T_{c,i}$ (K)	$v_{c,i} \times 10^6$ (m ³ mol ⁻¹)	ω_i
CO	35.0 [†]	132.9	93.1	0.049 [‡]
CO ₂	73.8 [†]	304.2	94.0	0.255 [‡]
H ₂ (effective)	20.5 [‡]	43.6	51.5	0
H ₂ O	220.5 [†]	647.3	56.0	0.344 [‡]
CH ₃ OH	81.0 [†]	512.6	118.0	0.572 [§]

[†]From Reid *et al.* (1977).

[‡]From Prausnitz (1969).

[§]From Tsionopoulos (1974).

validity. The most important disadvantage of the virial equation is that it can be used at low or moderate pressures only. For complicated mixtures like the one we are dealing with, the virial equation has to be truncated after the second virial coefficient. As a rough rule, the virial equation truncated after the second virial coefficient is valid up to pressures as calculated by eq. (16) (Prausnitz *et al.*, 1967):

$$P \leq \frac{T}{2} \frac{\sum_i y_i P_{c,i}}{\sum_i y_i T_{c,i}} \quad (16)$$

Equation (16) yields maximum pressures between 50 and 80 bar (depending on the temperature and composition) for actual methanol synthesis gas mixtures.

Second virial coefficients are calculated (Tsonopoulos, 1974) using the critical values and acentric factors from Table 2. In this way, the calculated second virial coefficients show fairly good agreement with the experimental ones (Dymond and Smith, 1969). Again no interaction coefficients are used in the calculation procedure for the cross-second virial coefficients. The second virial coefficient of the mixture is calculated as follows:

$$B_M = \sum_i \sum_j y_i y_j B_{ij} \quad (17)$$

3. LITERATURE RESULTS

3.1. Chemical equilibrium constants

Literature data on the methanol equilibrium show a great deal of scatter, while data on the water-gas shift equilibrium are scarce in the temperature range 200–300°C.

Some authors give relationships for K_p° as a function of temperature. These relationships are based on thermochemical data assuming ideal gas behaviour.

For comparison purposes, values of the chemical equilibrium constants at various temperatures, calculated from these relationships, are listed in Tables 3 and 4.

In Figs 1 and 2, the experimental K_p values of several authors are represented. The K_{p1} values were obtained at pressures between 1 and 200 bar, the K_{p2} values all at 1 bar. For comparison, some of the data of Tables 3 and 4 are represented in Figs 1 and 2 as well.

3.2. Correction for non-ideal gas behaviour

Very little information about deviations from ideal gas behaviour is available for the methanol synthesis

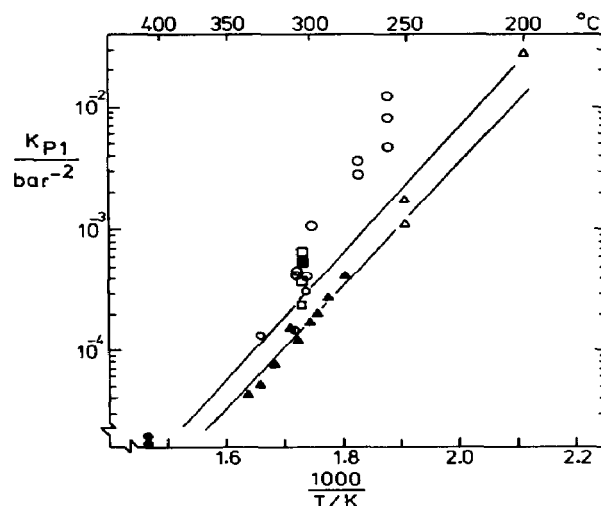


Fig. 1. Literature data on the chemical equilibrium of the methanol reaction. O, Von Wettberg and Dodge (1930); ●, Brown and Galloway (1928); □, Smith and Hirst (1930); ■, Smith and Branting (1929); △, Lacy *et al.* (1930); ▲, Newitt *et al.* (1928); upper line, Cherednichenko (1953); lower line, Zeise (1954).

Table 3. K_{p1}° values of several authors

Authors		$10^2 \times K_{p1}^\circ$ (200°C) (bar ⁻²)	$10^3 \times K_{p1}^\circ$ (250°C) (bar ⁻²)	$10^4 \times K_{p1}^\circ$ (300°C) (bar ⁻²)
Cherednichenko	(1953)	2.58	2.34	3.14
Thomas and Portalski	(1958)	1.96	1.83	2.52
Newton and Dodge	(1934)	2.13	2.20	3.20
Ewell†	(1940)	1.74	1.67	2.38
Zeise†	(1954)	1.33	1.30	1.92

† Author does not give relationships; values are calculated by interpolation if necessary.

Table 4. K_{p2}° values of several authors

Authors		$10^3 \times K_{p2}^\circ$ (200°C)	$10^2 \times K_{p2}^\circ$ (250°C)	$10^2 \times K_{p2}^\circ$ (300°C)
Hahn	(1903)	2.80	0.76	1.74
Singh and Saraf	(1977)	4.18	1.10	2.43
Moe	(1962)	4.77	1.20	2.58
Zeise†	(1954)	4.83	1.18	2.47

† Values are calculated by interpolation.

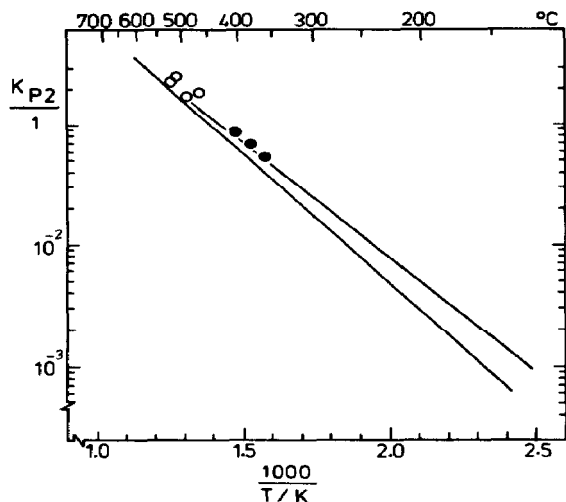


Fig. 2. Literature data on the chemical equilibrium of the water-gas shift reaction. O, Oki and Mesaki (1973); ●, Podolski and Kim (1974); upper line, Zeise (1954); lower line, Hahn (1903).

mixture. Ewell (1940) calculated the fugacity coefficients for carbon monoxide, hydrogen and methanol using Lewis and Randall's rule (Lewis and Randall, 1923):

$$\varphi_i(\text{mixture}) = \varphi_i(\text{pure component, same } P, T). \quad (18)$$

The result of this simplification is that the fugacity coefficients become a function of pressure and temperature only and not of composition.

4. EXPERIMENTAL

4.1. Equipment

In order to determine the equilibrium constants of reactions (1) and (2), a cylindrical fixed-bed reactor packed with catalyst particles (Haldor Topsoe MK

101) was used. The mean particle diameter was 0.45 mm. The inner diameter of the reactor was 4.6 mm, the outer diameter 6.4 mm and the length 2.5 m. The amount of catalyst used was 5.5×10^{-2} kg.

A flow scheme of the equipment is presented in Fig. 3. The feeds, consisting of mixtures of carbon monoxide, carbon dioxide and hydrogen, were drawn from gas cylinders (1). The pressure in the reactor was adjusted with a pressure reducer (2). The reactor (4) was heated in a liquid metal bath (5), thermostated by a proportional thermal controller. Before entering the reactor, the gas was passed through a preheater (3). A detailed sketch of the reactor is given in Fig. 4. In order to avoid local temperature differences, a metal cylinder was placed between the heating coil and the reactor in the liquid metal bath. This cylinder consisted of three layers: successively stainless steel, copper and stainless steel. In this way the temperature differences in the reactor were limited to less than 0.3°C . Temperature measurements were carried out with an accurately calibrated thermowell (6) placed in the exit stream of the reactor. The pressure of the inlet stream of the

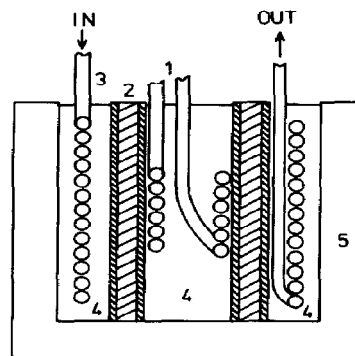


Fig. 4. Reactor set-up for the determination of the chemical equilibria of the methanol- and the water-gas-shift equilibrium. 1, Heating coil; 2, cylinder: ▨, stainless steel; ▤, copper; 3, reactor-tube; 4, liquid metal; 5, liquid metal bath.

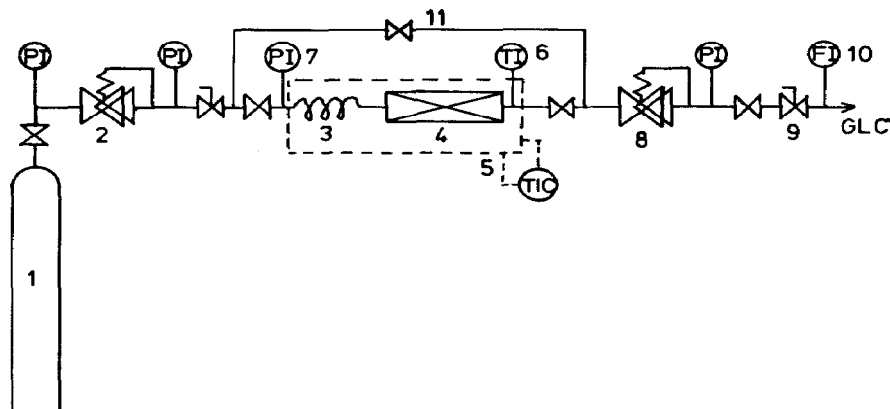


Fig. 3. Flow scheme of the equipment used for the determination of the methanol equilibrium and the water-gas shift equilibrium. 1, Gas cylinder; 2, pressure reducer; 3, preheater; 4, reactor; 5, thermostat; 6, thermowell; 7, manometer; 8, pressure reducer; 9, needle valve; 10, soap bubble meter; 11, bypass.

reactor was measured with an accurate manometer (7). The gas flow through the reactor could be adjusted with a needle valve (9). In order to obtain a constant pressure drop over this needle valve, a second pressure reducer (8) was used. The gas flow was measured with a soap bubble meter (10). The product lines were heated electrically to avoid condensation of methanol and water. The gas flow was connected to GLC apparatus equipped with an automatic vapour sampling valve. The feed could be sampled separately through a reactor bypass (11).

4.2. Analysis

A schematic drawing of the GLC apparatus used is given in Fig. 5. Gas samples of 1 ml were injected. The column temperature as well as the sampling valve temperature was maintained at 100°C.

Helium was used as the carrier gas. A switching valve was installed at the end of column A (2 mm i.d., 0.5 m long, packed with Porapak Q). The switching time was chosen in such a way that carbon monoxide, carbon dioxide and water were separated in column B (2 mm i.d., 6 m long, packed with Porapak Q), while methanol and organic by-products (except methane and ethane) were separated in column C (2 mm i.d., 6 m long, packed with Carbowax 1540 on Chromosorb WAW). The components of column B were detected with a TCD, those of column C with an FID. Calibrations of the detectors were carried out regularly in order to assure accurate analysis. Hydrogen was not determined directly in the analysis, but from a material balance:

$$y_{H_2} = 1 - \sum_i y_i \quad (i \neq H_2). \quad (19)$$

4.3. Measurements

The measurements were always carried out under steady-state conditions. In order to check if equilibrium had been reached within the experimental accuracy, different feed rates were used during one experiment (same P, T and feed composition). No problems were encountered in reaching equilibrium. However, when too small feed rates were used, less accurate results were obtained. This was due to the occurrence of side reactions: formation of dimethyl ether and hydrocarbons, the latter probably formed on the reactor wall (stainless steel). For each experiment the material balances over the reactor for hydrogen, carbon and oxygen were calculated.

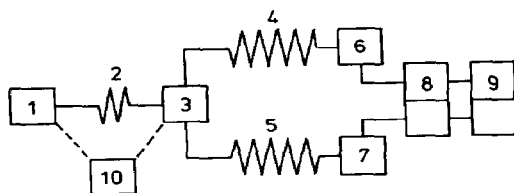


Fig. 5. GLC apparatus used. 1, Sampling valve; 2, column A; 3, switching valve; 4, column B; 5, column C; 6, TDC; 7, FID; 8, integrator; 9, recorder; 10, timer.

The deviations in these material balances were less than 1% in most cases. Measurements were carried out at pressures between 10 and 80 bar and temperatures between 200 and 270°C. Inlet concentrations were varied as follows: $y_{CO} = 0.1$ –0.2; $y_{CO_2} = 0.03$ –0.1. Outlet concentrations of methanol and water were varied as follows: $y_{CH_3OH} = 0.01$ –0.15; $y_{H_2O} = 0.003$ –0.03 depending on the experimental conditions.

4.4. Results

The results of 23 experiments are listed in Table 5. The values of the chemical equilibrium constants presented in the last two columns were calculated from the experimental results using the following equations:

$$K_{p_1} = \left(\frac{y_{CH_3OH}}{y_{CO} y_{H_2}^2} \right)_{Eq} \frac{1}{P^2} \quad (20)$$

$$K_{p_2} = \left(\frac{y_{H_2O} y_{CO}}{y_{CO_2} y_{H_2}} \right)_{Eq} \quad (21)$$

The chemical equilibrium constants can be calculated from eqs (11) and (12) using successively one of the equations of state listed in Table 1 to calculate K_ϕ . In these calculations we use the experimental values of the mole fractions, the temperature and the pressure.

For each equation of state the relative variance between the calculated and experimental equilibrium constants was calculated from eq. (22):

$$S^2 = \frac{1}{N} \sum_{i=1}^N \left(\frac{K_{p_{calc}} - K_{p_{exp}}}{K_{p_{exp}}} \right)^2 \quad (22)$$

The relative variances are given in Table 6.

Table 5. Results of the chemical equilibrium experiments

Run	P (bar)	T (K)	$10^3 \times K_{p_1}$ (bar $^{-2}$)	$10^3 \times K_{p_2}$
1	10.7	474.4	16.3	4.10
2	19.3	488.2	8.10	6.24
3	10.6	489.6	7.48	5.46
4	10.7	502.3	4.07	7.50
5	9.8	509.3	2.95	8.75
6	10.8	530.4	1.17	13.3
7	11.5	532.3	1.08	13.4
8	26.5	506.4	3.40	9.41
9	34.6	509.1	3.01	9.16
10	32.6	515.2	2.25	10.3
11	25.2	518.7	1.91	11.1
12	23.8	534.9	1.04	15.4
13	26.5	538.1	0.892	15.3
14	44.5	472.4	18.6	4.52
15	63.1	481.0	12.3	6.10
16	48.7	487.8	8.43	6.53
17	50.7	488.1	8.62	6.19
18	71.6	495.2	6.06	8.06
19	50.5	502.3	4.23	8.05
20	72.2	509.3	3.15	10.3
21	44.2	516.9	2.11	10.6
22	43.4	530.0	1.22	13.7
23	49.8	536.2	0.970	14.8

Table 6. Relative variances between the calculated and experimental equilibrium constants

Equation of state used	$10^3 \times S_1^2$ (methanol equilibrium)	$10^3 \times S_2^2$ (water-gas shift equilibrium)
IGL	1.40	5.17
RK	4.54	3.47
RKS	0.64	3.47
PR	1.44	3.25
Vir	7.19	5.05

These variances are due to experimental inaccuracies and to "lack of fit". An accuracy study of our experiments, which is not given here, shows that the experimental inaccuracy in K_{p_2} ($\pm 5\%$) is much larger than that in K_{p_1} ($\pm 1\%$). This can be traced in Table 6.

The S_2^2 values do not differ very much, indicating that the experimental inaccuracies are larger than the lack of fit. The S_1^2 values, however, differ a great deal and the largest S_1^2 values are probably caused by a lack of fit of the equation of state used. Therefore conclusions about the suitability of an equation of state can be drawn only from the results of the methanol equilibrium.

A striking result is that the ideal gas law gives better results than the Redlich-Kwong equation of state and the virial equation truncated after the second virial coefficient. However, the best results are obtained using the Soave-Redlich-Kwong equation of state. In Figs 6 and 7 the experimental equilibrium constants corrected to a pressure of 1 bar with the use of the Soave-Redlich-Kwong equation of state, are given, as well as eqs (11) and (12). It can be seen that there is good agreement between the (semi-)experimental points and

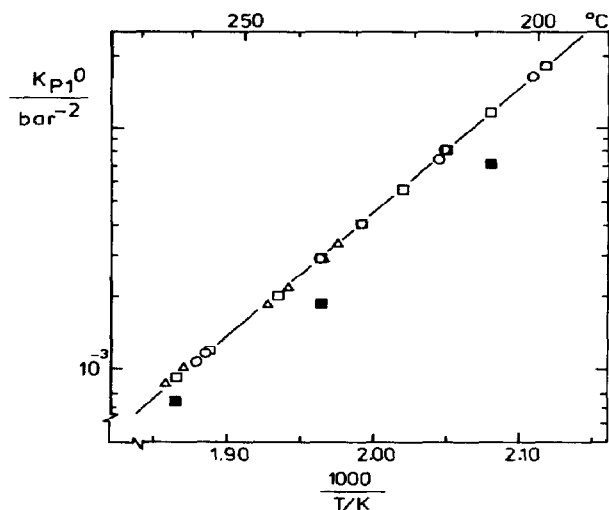


Fig. 6. Results of the chemical equilibrium of the methanol reaction. Open symbols: experimental points corrected to 1 bar with the use of the Soave-Redlich-Kwong equation of state. $\circ$, $P < 20$ bar; $\triangle$, $20 < P < 40$ bar; $\square$, $P > 40$ bar; $\blacksquare$; experimental points ($P > 40$ bar) corrected with Lewis and Randall's rule.

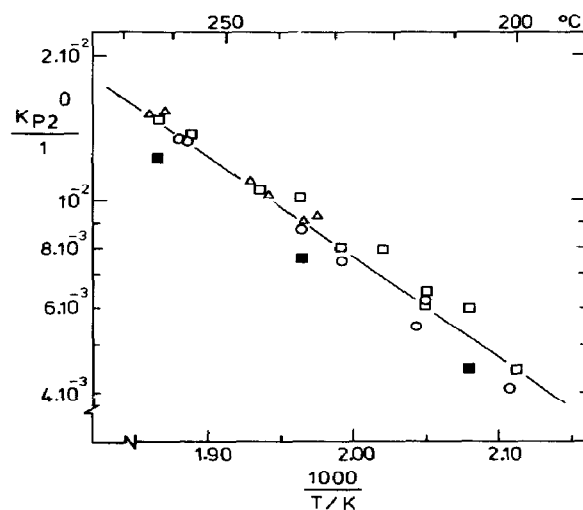


Fig. 7. Results of the chemical equilibrium of the water-gas shift reaction. Open symbols: experimental points corrected to 1 bar with the use of the Soave-Redlich-Kwong equation of state. $\circ$, $P < 20$ bar; $\triangle$, $20 < P < 40$ bar; $\square$, $P > 40$ bar; $\blacksquare$, experimental points ($P > 40$ bar) corrected with Lewis and Randall's rule.

the relationships derived from thermochemical data assuming ideal gas behaviour. For comparison purposes, experiments 15, 20 and 23 are also corrected to a pressure of 1 bar using Lewis and Randall's rule (Lewis and Randall, 1923). As can be clearly seen, this leads to serious errors.

Comparison of our results with the literature data shows that our results for the methanol equilibrium are in fairly good agreement with the results of Ewell (1940). This can be seen in Table 7.

Extrapolating our results for the water-gas shift equilibrium to temperatures between 350 and 550°C shows that our results are in agreement with the experimental values of Oki and Mezaki (1973) and those of Podolski and Kim (1974). This can be seen in Table 8.

5. CONCLUSIONS

The chemical equilibria of the methanol reaction and the water-gas shift reaction can be described very well by relationships derived from thermochemical data assuming ideal gas behaviour and correcting for non-ideal gas behaviour by using the Soave-Redlich-Kwong equation of state (Soave, 1971).

Our results for the methanol equilibrium (at 1 bar) show fairly good agreement with the results of Ewell (1940). The experimentally determined K_{p_2} values of Oki and Mezaki (1973) and of Podolski and Kim (1974) are in agreement with our results, although their results were obtained between 350 and 550°C.

Correcting for non-ideality by using the Soave-Redlich-Kwong equation of state (Soave, 1971) gave significantly better results than a correction based on the original Redlich-Kwong equation of state

Table 7. Comparison of the literature data on the methanol equilibrium with our results

Ref.	$10^2 \times K_{p_1}^o$ (200°C) (bar ⁻²)	$10^3 \times K_{p_1}^o$ (250°C) (bar ⁻²)	$10^4 \times K_{p_1}^o$ (300°C) (bar ⁻²)
Ewell (1940)	1.74	1.67	2.38
This work	1.74	1.59	2.21

Table 8. Comparison of the literature data on the water-gas shift equilibrium with our results

Ref.	<i>T</i> (°C)	$10^1 \times K_{p_2}^o$	$10^1 \times K_{p_2}^o$ (this work)
Oki and Mezaki (1973)	467	1.9	1.7
	489	1.8	2.0
	522	2.3	2.6
	511	2.6	2.4
Podolski and Kim (1974)	360	0.55	0.57
	381	0.70	0.72
	404	0.89	0.93

(Redlich and Kwong, 1949), the Peng–Robinson equation of state (Peng and Robinson, 1976), the virial equation of state truncated after the second virial coefficient, Lewis and Randall's rule (Lewis and Randall, 1923), or not correcting at all for non-ideality, thus assuming ideal gas behaviour. Furthermore, it has to be emphasized that assuming ideal gas behaviour gives much better results than Lewis and Randall's rule.

Acknowledgements—We thank Haldor Topsoe for delivering their methanol synthesis catalyst MK 101 and Gasunie for delivering calibration gas mixtures.

NOTATION

Abbreviations

IGL	ideal gas law
PR	Peng–Robinson equation of state
RK	Redlich–Kwong equation of state
RKS	Soave–Redlich–Kwong equation of state
Vir	virial equation

Symbols

<i>a</i>	constant in PR, RK and RKS
<i>a</i> ₁ – <i>a</i> ₆	constants in K_p^o equation
<i>b</i>	constant in PR, RK and RKS, m ³ mol ⁻¹
<i>b</i> ₁ – <i>b</i> ₂	constants in K_p^o equation
<i>B</i>	second virial coefficient, m ³ mol ⁻¹
<i>C</i>	third virial coefficient, m ⁶ mol ⁻²
<i>f</i>	partial fugacity, bar
<i>K_f</i>	equilibrium constant based on partial fugacities
<i>K_p</i>	equilibrium constant based on partial pressures
<i>K_φ</i>	ratio of fugacity coefficients
<i>n</i>	quantity, mol
<i>N</i>	number of experiments

<i>p</i>	partial pressure, bar
<i>P</i>	total pressure, bar
<i>R</i>	gas constant, J mol ⁻¹ K ⁻¹
<i>S</i> ²	variance
<i>T</i>	temperature, K
<i>v</i>	molar volume, m ³ mol ⁻¹
<i>V</i>	volume, m ³
<i>y</i>	mole fraction
<i>Z</i>	compressibility factor

Greek letters

ΔC_p	specific heat change, J mol ⁻¹ K ⁻¹
ΔH	enthalpy change, J mol ⁻¹
ΔG	Gibbs energy change, J mol ⁻¹
ϕ	fugacity coefficient
ω	acentric factor

Subscripts

<i>c</i>	at the critical point
calc	calculated
Eq	at equilibrium
exp	experimental
<i>f</i>	of formation
<i>i, j</i>	indicates component
<i>M</i>	mixture
ref	reference
<i>o</i>	feed conditions
1	methanol reaction
2	water reaction

Superscript

<i>o</i>	indicates standard pressure (1 bar)
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